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EVENT AND DATE

LECTURE 2 – PLM (PRODUCT & SERVICES DEVELOPMENT VIEW)

CS39A0040 Product & Service Development

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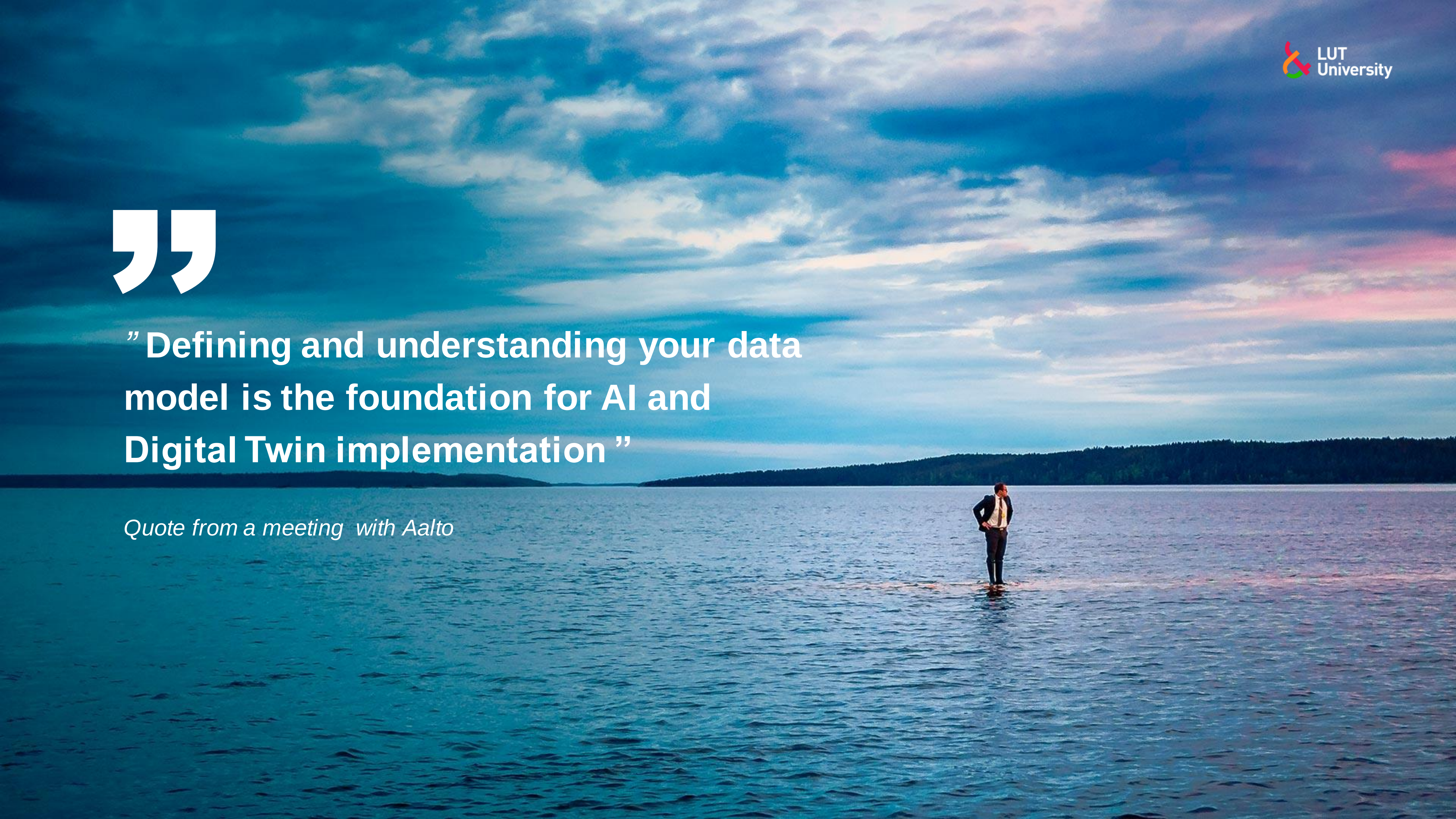
AGENDA FOR LECTURE 2

- Product Lifecycle Management (PLM)
- Group Assignment Plan Q&A



”Defining and understanding your data model is the foundation for AI and Digital Twin implementation ”

Quote from a meeting with Aalto

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INTRODUCTION TO PLM AND PRODUCT & SERVICE DEVELOPMENT

What is the difference?


PORSCHE
911
S GO 4001
PORSCHE MUSEUM
PORSCHE
911 Carrera 4S
S GO 4002

PRODUCT LIFECYCLE

Platform and logical architecture remain.

Technology and the product changes over the generations.

Lifecycle management essential throughout Customer eXperience (CX).

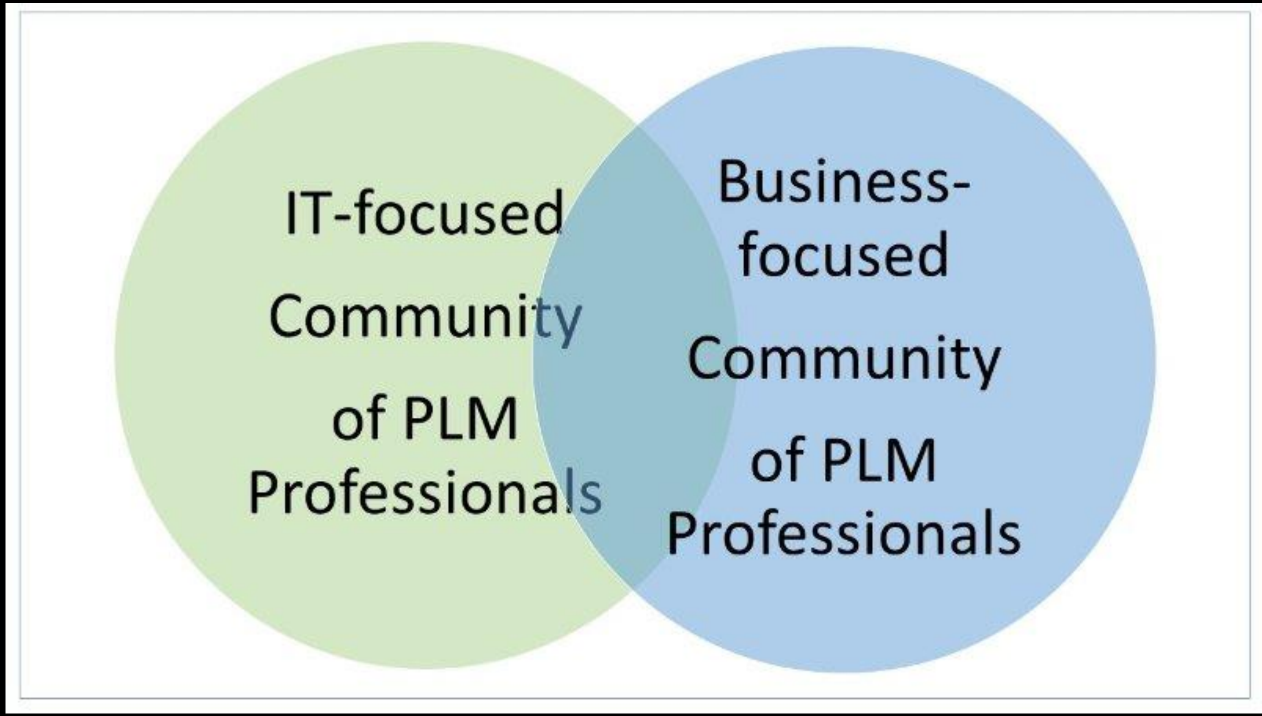
LIFECYCLE IMPORTANCE

Complex systems are dependent on their modules lifecycle.

Criticality of the modules define the systems service life.

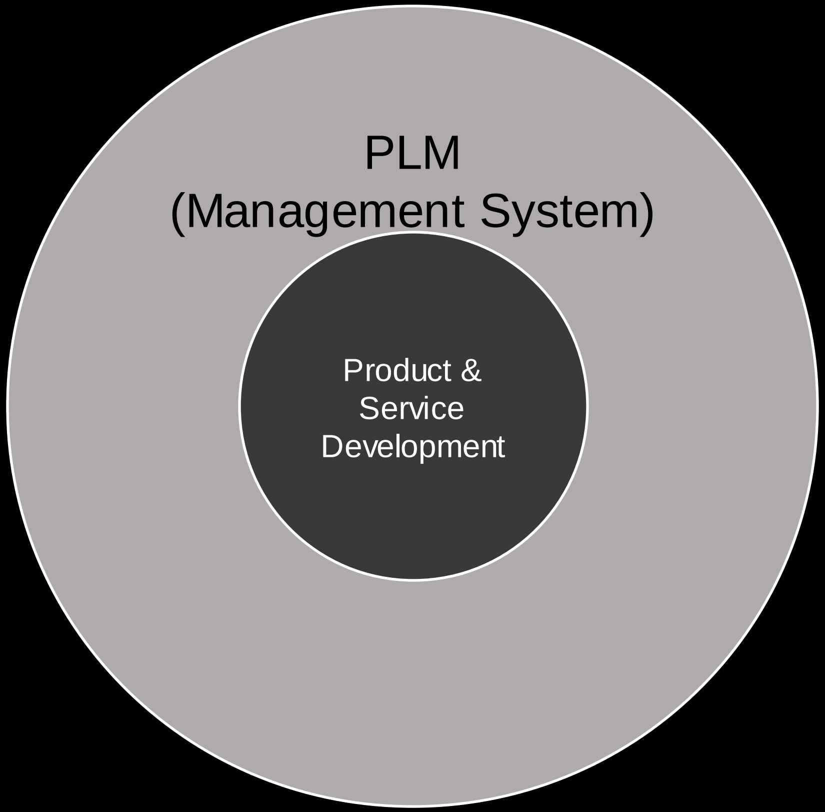
Service lifecycle is part of PLM.

PRODUCT LIFECYCLE MANAGEMENT'S TWO COMMUNITIES

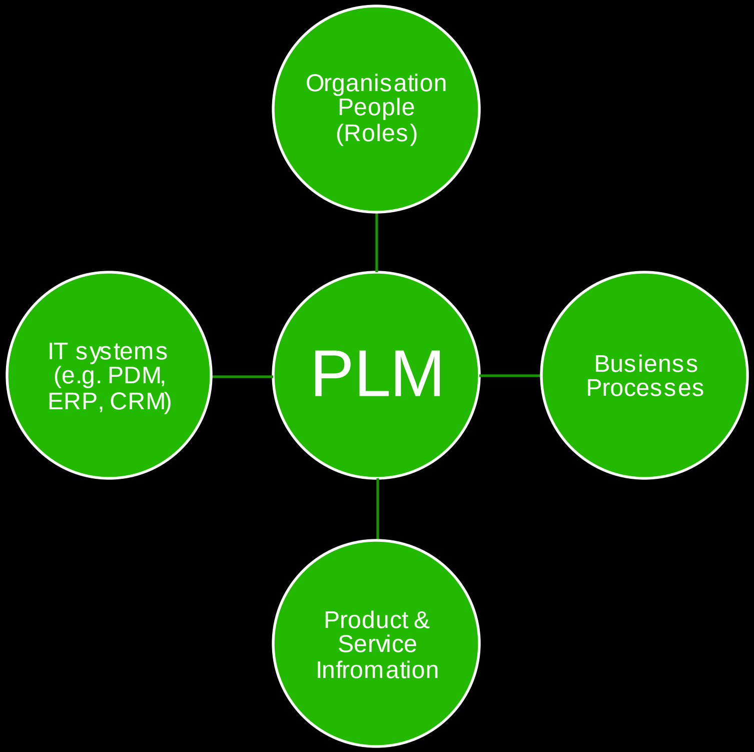


John Stark, 2015

PRODUCT LIFECYCLE MANAGEMENT IS THE MANAGEMENT SYSTEM



PLM elements



PLM DEFINITIONS

Product Lifecycle Management (PLM)

- integrative information-driven ***business concept*** comprised of **people**, **processes**/**practices**, **and technology** to all aspects of a product's life and its environment, from its design through manufacture, deployment and maintenance – culminating in the product's removal from service and final disposal.

(Grieves 2010).

PDM and PLM IT systems

- based on data model, enable accessing, updating, manipulating and reasoning about product information that is being produced in a fragmented and distributed environment.

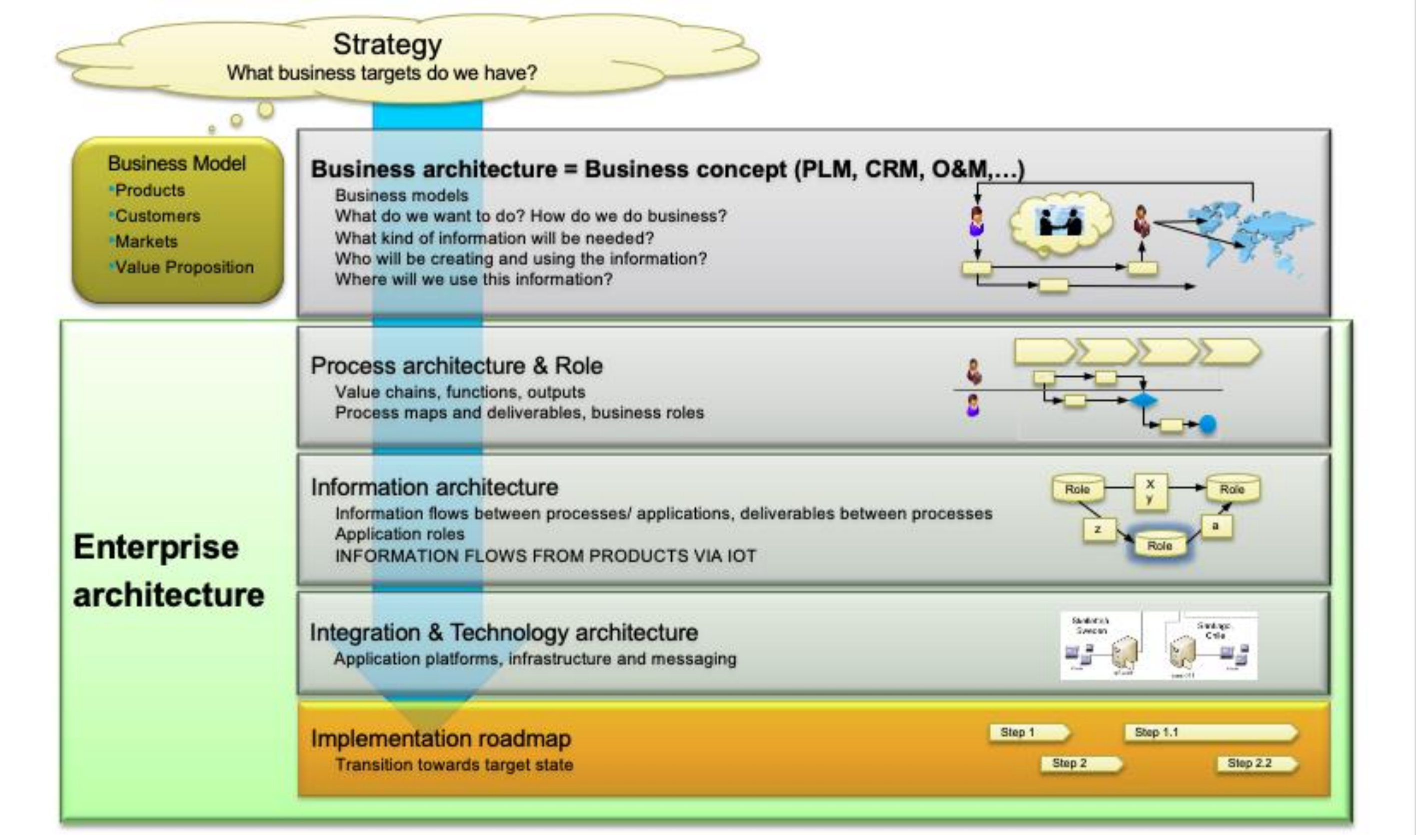
(Saaksvuori 2008)

PLM DEFINITIONS

- PLM is a strategic concept (business approach)to manage the IP of the product overit’s whole lifecycle.
- The concept includes business elements of a company as processes (especially change and configuration management processes), organization structures, methods and supporting IT systems. Emphasis has to be laid on the Human Factors.
- All areas of a company that are connected to the product as well as the related processes and resources are included also

Fachbach: Future PLM, 2011

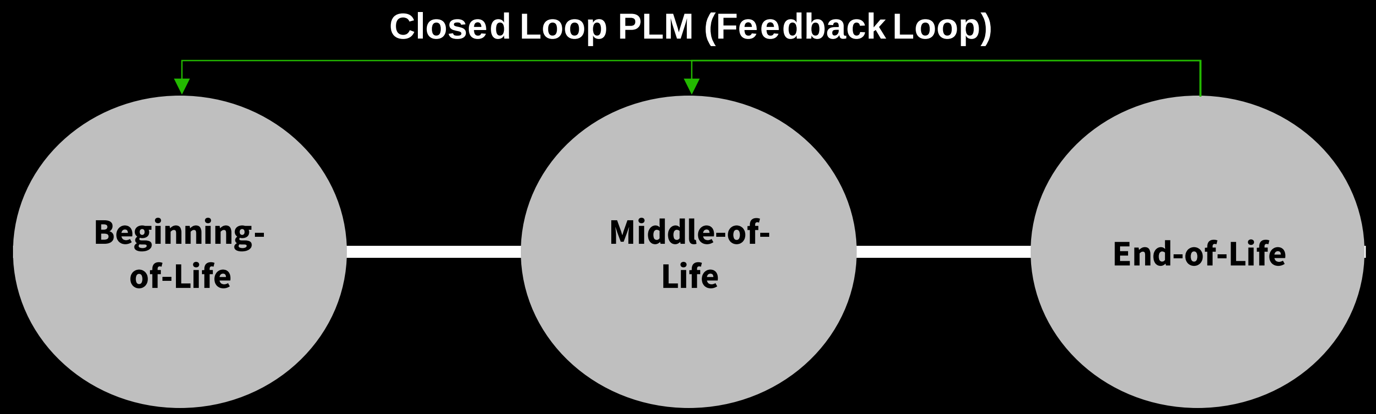
PLM IMPACT ON THE ENTERPRISE ARCHITECTURE



- PLM goal come from strategy impacting both iPSS and system.
- Part of the busiens architecture with the iPSS at the core.
- Defines & impacts the enterprise architecture across all core and supporting business processes.

Donoghue et al., 2018

PRODUCT LIFECYCLE



Activities:
- Design
- Manufacturing

Focus:
- Manufacturer & Market

Occusrance:
- Virtual
- (Semi-)physical

Activities:
- Delivery
- Operations
- Support

Focus:
- Customer

Occurance
- Virtual & physical alignment

Activities:
- Upgrades & modernisations
- Last Time Buy

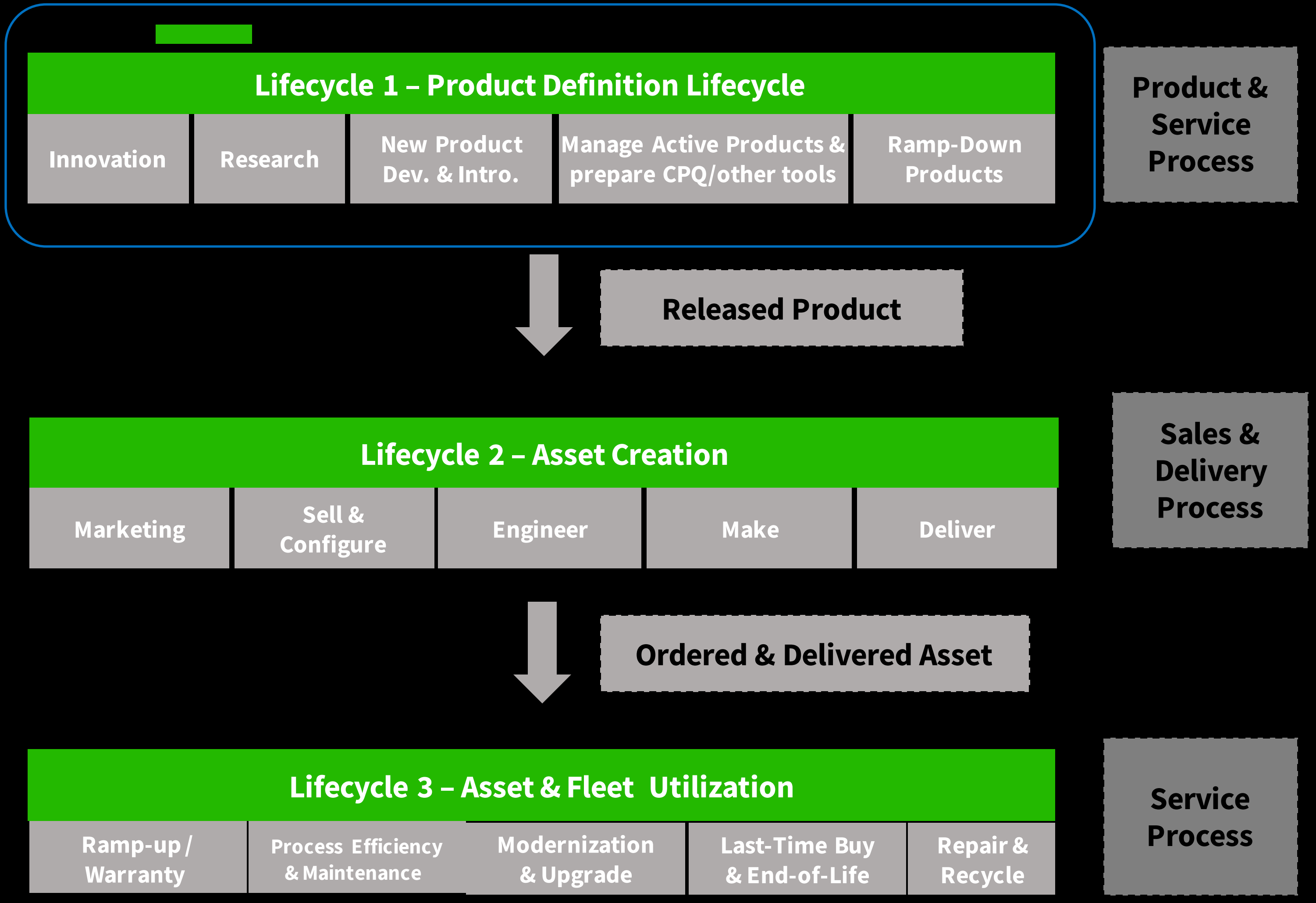
Focus:
- Customer (Service)

Occurance:
- Physical & virtual (misalingment)

Terzi, et al., 2010

THE CORE IS TO DEVELOP, OFFER AND MAINTAIN YOUR PRODUCTS & SERVICES OVER THE CLM

THE DIGITAL THREAD ENABLES CUSTOMER & PRODUCT LIFECYCLE MANAGEMENT

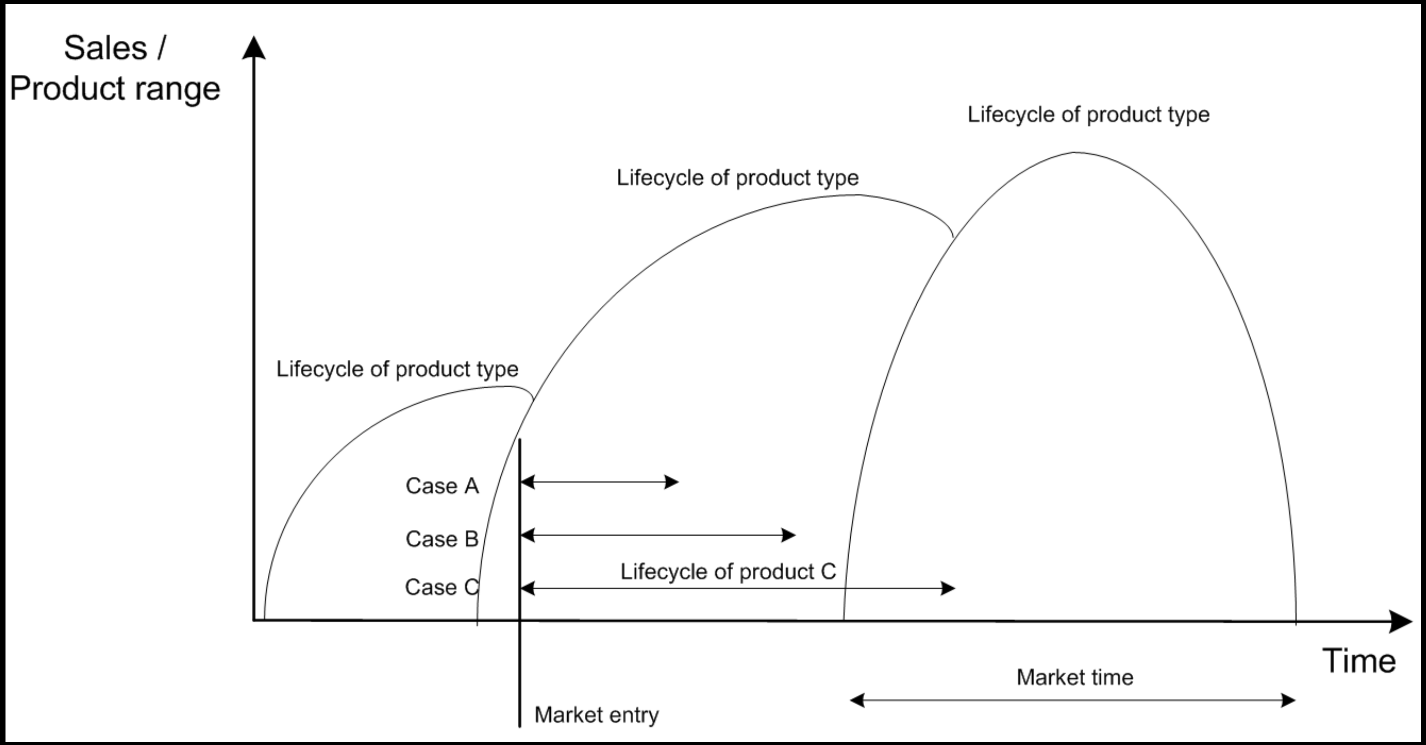


Lifecycle 1 defines and manages the product definition ensuring the product 360, and builds the capability for Lifecycle 2 Asset Creation (sales operation).

Lifecycle 2 manages the marketing, sales and CPQ process that use the product 360 information to generate customer specific configured and priced solutions that are handed to delivery & invoicing.

Lifecycle 3 manages the installed base and each asset information (SBOM) to ensure the assets performance with offered services.

COUPLING OF PRODUCT TYPE AND INSTANCE



Terzi et al (2010) defined three lifecycles:

1. Beginning-of-life (design)
2. Middle-of-life (marketing, sales, & delivery)
3. End-of-life (retirement from use & recycle elements)

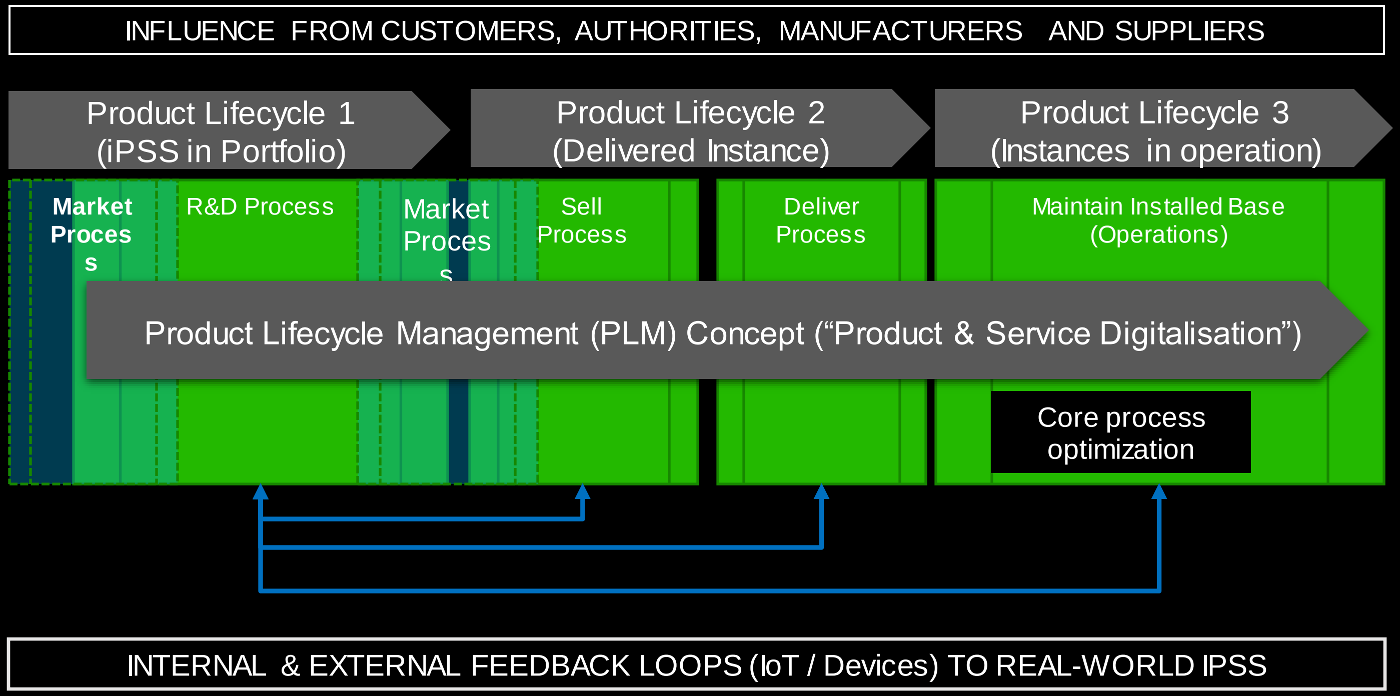
Does not clarify or over-simplifies the operational lifecycle.

Portfolio view from all “instances & offering” or a in use instance lifecycle view.

Lifecycle extension and market time are dependent on services.

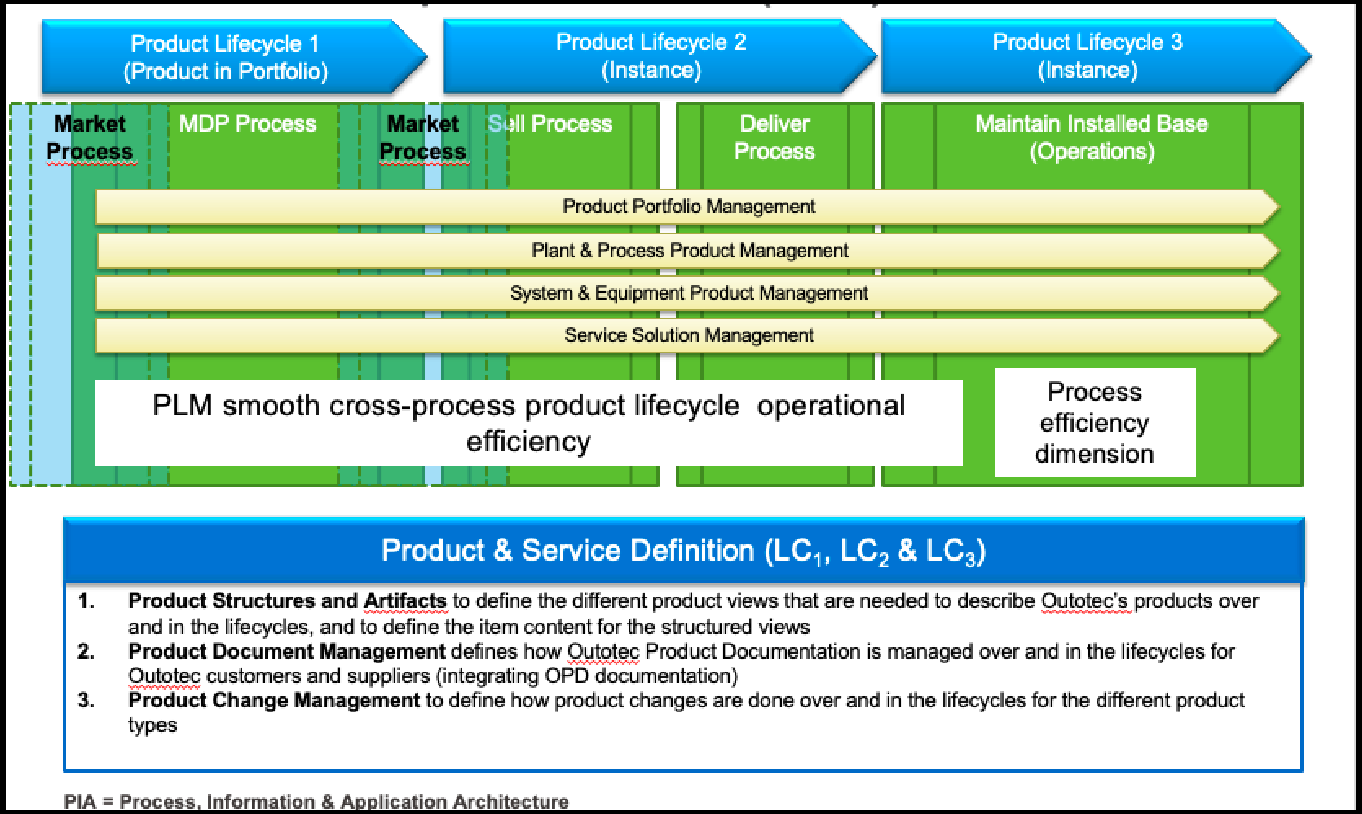
Papinniemi & Donoghue, 2019

PRODUCT LIFECYCLE MANAGEMENT CONCEPT



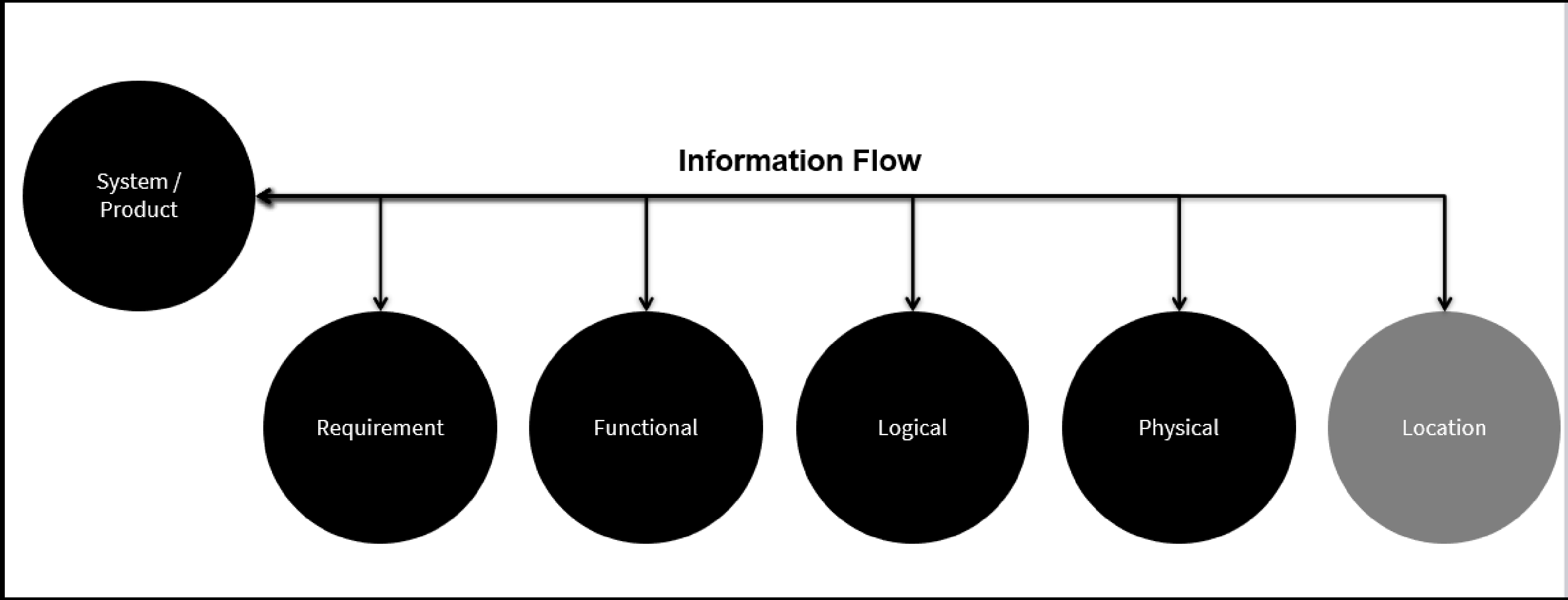
Donoghue, 2017

PLM CONCEPT & ITS ELEMENTS (PIA)



Donoghue, 2018

REVISED PLM RFLP STRUCTURE

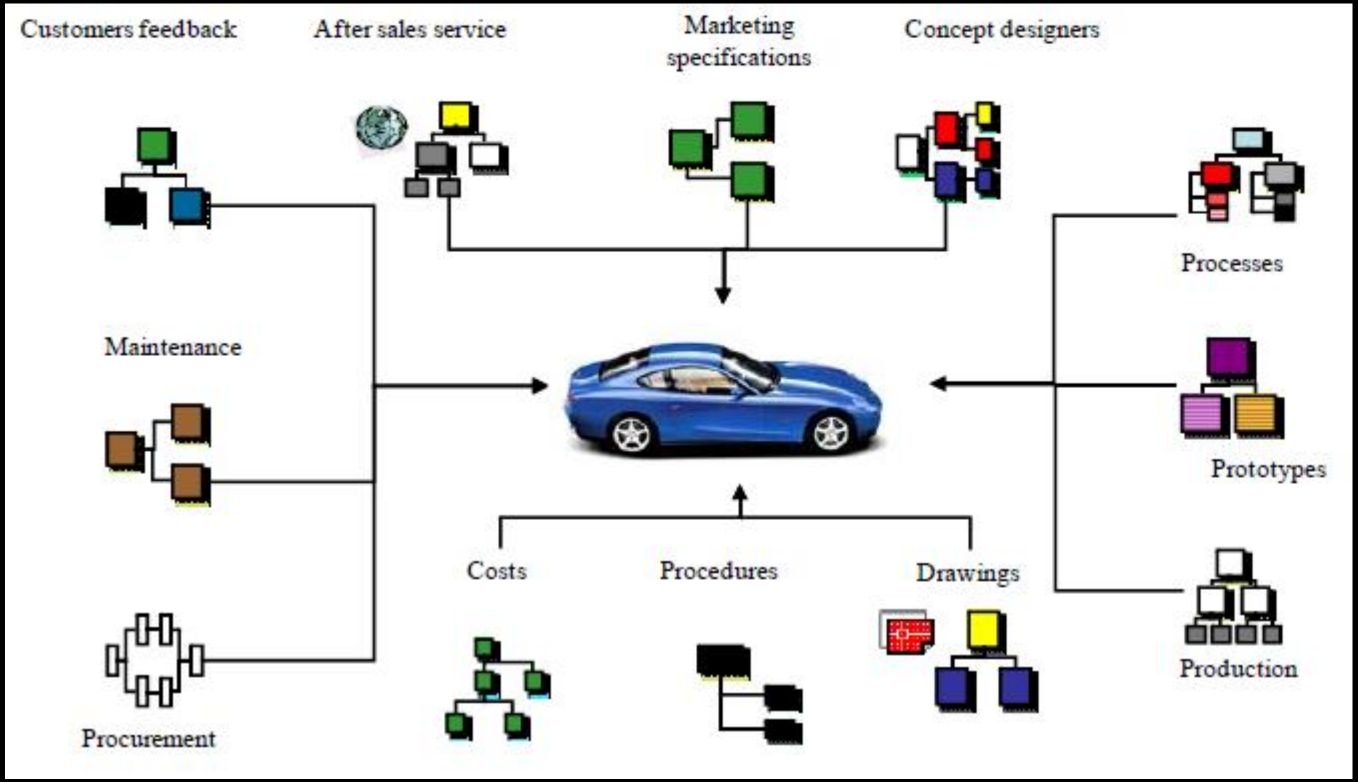


The Product & Services can be defined from multiple behavioral viewpoints that need to be managed over the Lifecycle.

R: Requirement
F: Functional
L: Logical
P: Physical
L: Location

Donoghue, 2021

EXTENDED PRODUCT STRUCUTRES VIEWS FOR A CAR



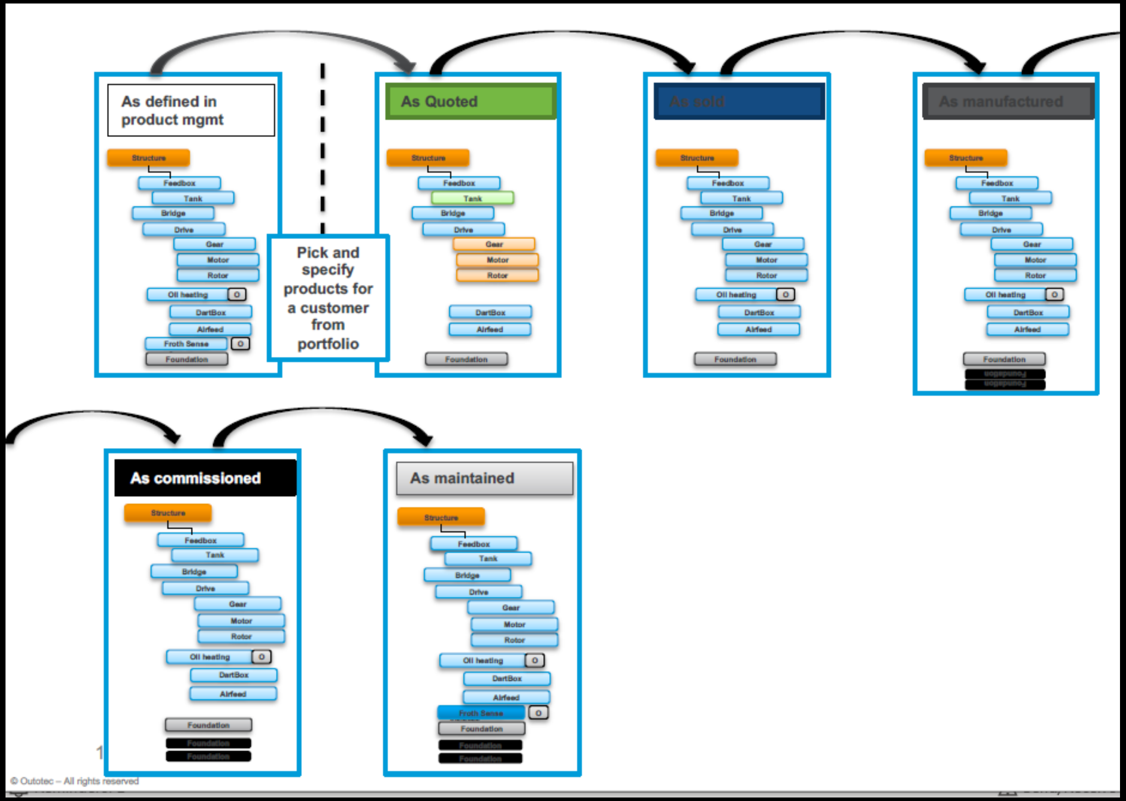
Additional information structures exist to give extended insight to the iPSS

New structures should be added as understanding and need increases.

Can AI enable this information structures to developed automatically and feed requirement to SRD?

Source: Alberto Codrino, in Terzi et al.

PSS INFORMATION FLOW



- iPSS defined in R&D and SRD.
- Select iPSS from portfolio to offer to customer:
  - Define as quoted
  - Define as sold (proposed)
- Sold iPSS forms the starting point for delivery and commissioning (ETO vs. CTO)
- As-commissioned is the starting point for services.
- As-maintained information is the operational lifecycle information about the iPSS.

Change Management is used to track technical and business changes

Documentation, 3D models, Real-time Simulation FEM, CFD etc. are used to visualize the iPSS

Donoghue et al, 2016

INCREASED PRODUCT COMPETITIVENESS

Product Management deployment – Focus areas

Business Capability	**Role & Skill**	**Product content**	**Process**
• Business, Technology & Product strategies linked together guiding PM action planning • Coordinated processing of Continuous Product Improvements	• Role of product managers and lead design engineers are used • Productization understood and used • Local product management competence development is in place • Global product management development network is active • Support for product management and productization is in place • Roles of product organizations are in place	• First product(s) for each product line productized and defined according to framework Product portfolio is managed actively • Product Information is shared & used • Productization priority list & roadmap are managed according to PL's own capabilities	• Core process in place • Process & related instructions, decision making, process & project templates available, known & understood • R&D process related tools are in use • Local process owner has taken his role and is known to relevant teams

Key Performance Indicators (KPI): Operative product management

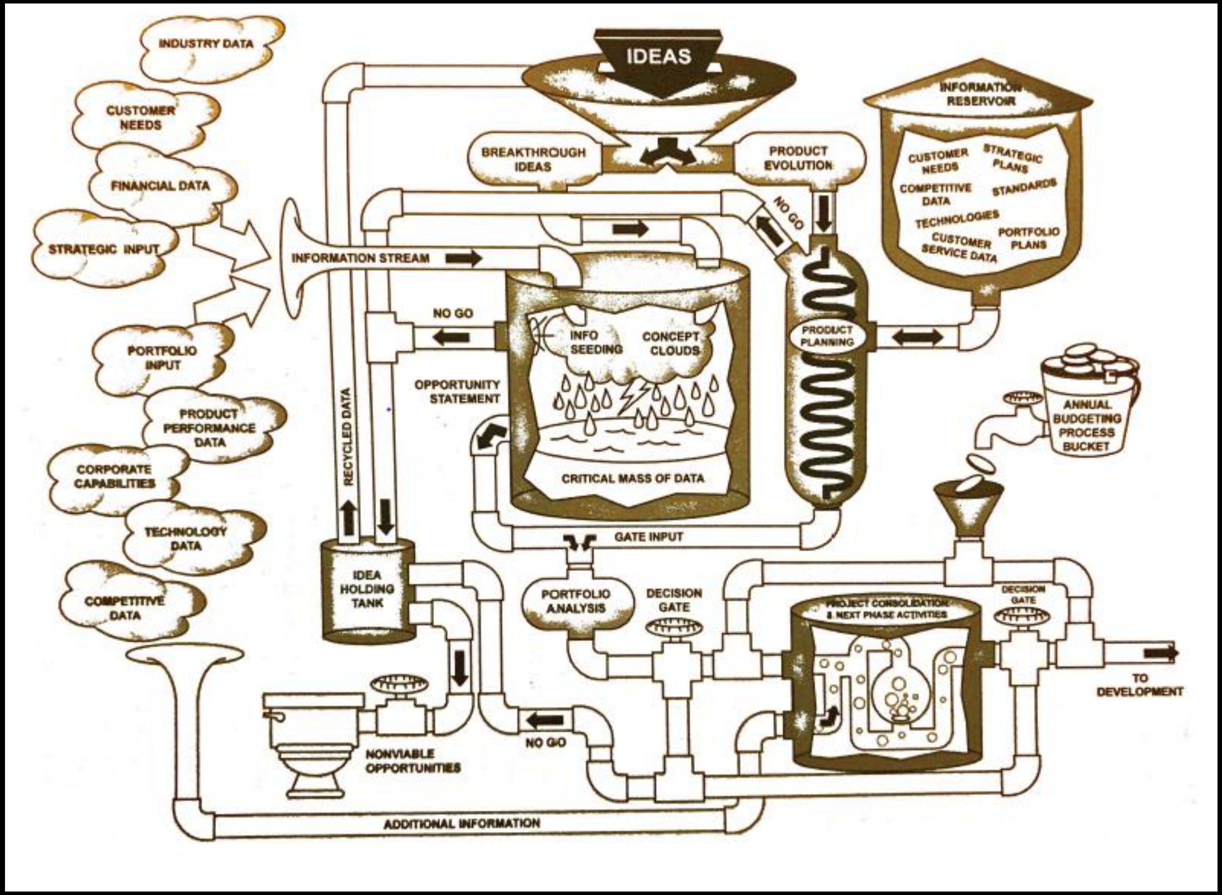
PLM LIMITATIONS

- Lifecycle view is often manufacturer centric
- PLM management is evolving for service and Product-Service Systems
- PLM should also be viewed from the customer and user perspctive
- Installed base management is challenging
- Connectivity and intelligence offers a way to manage the PLM.
- Data model and information flow is complex.

PRODUCT & SERVICE DEVELOPMENT VIEW

What approach to take depends on the product, service and system

NEW PRODUCT DEVELOPMENT AND INTRODUCTION

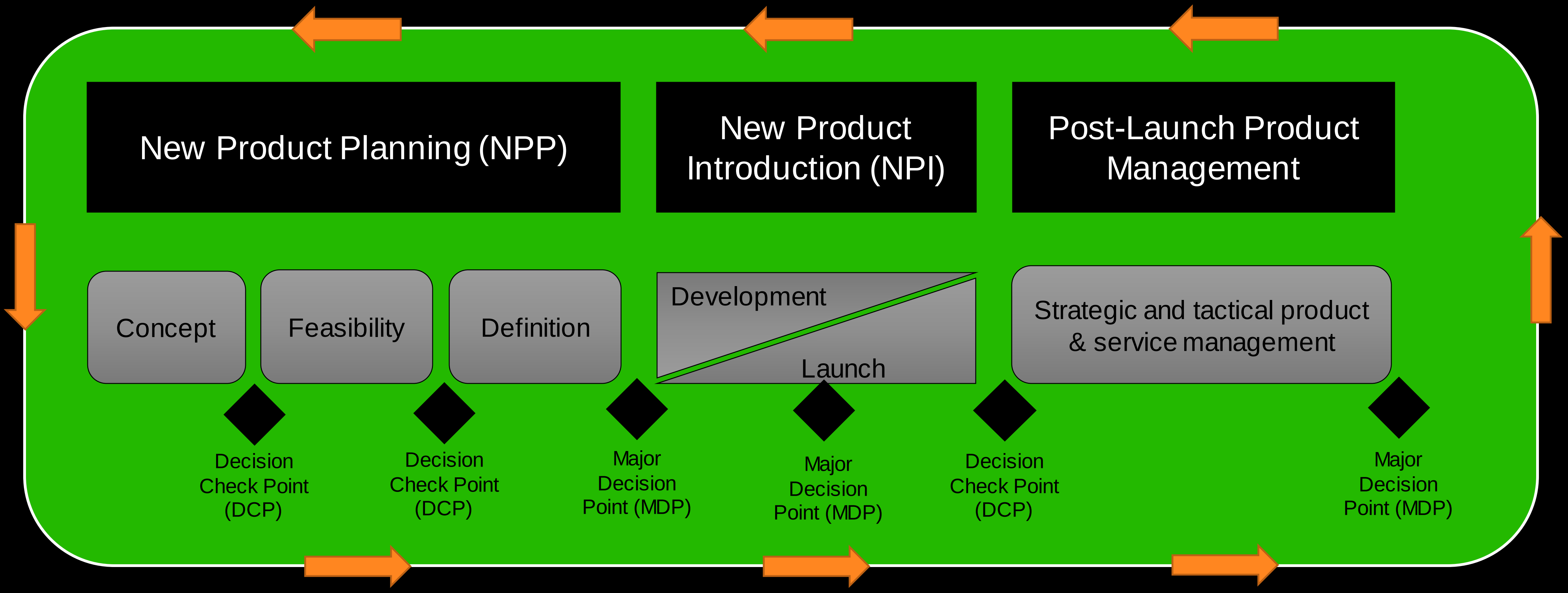


- Processes are linear and markets are not.
- Planning must be done with time to speed up in later phases

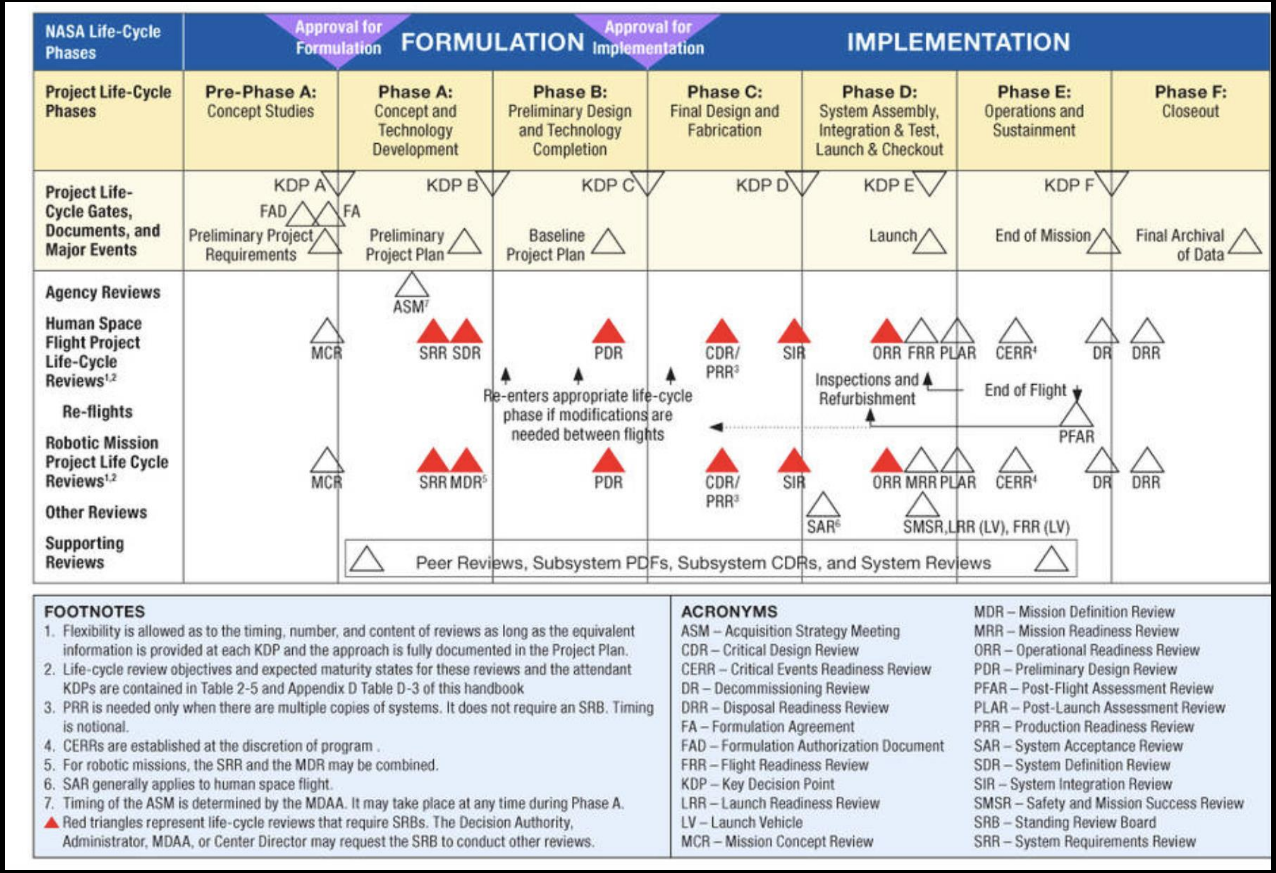
(Haines, 2007)

Product Management's role

GENERIC PLANNING & DEVELOPMENT PROCESS



DEVELOPMENT PROCESS – WATERFALL



Waterfall Development Model Example

- Oldest development model applied typically to physical products.
- Decision points define the progress (yes, no, more information).
- Rapid market changes difficult to include
- Examples:
  - Ships, cars, aerospace, defense, heavy industry.

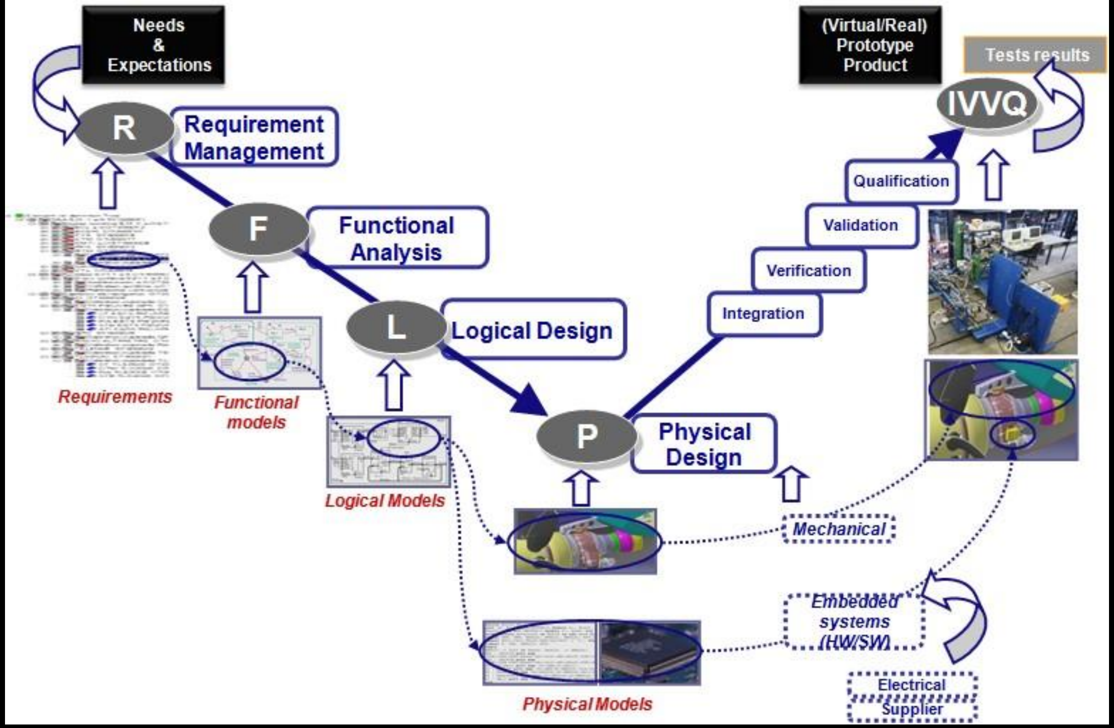
DEVELOPMENT PROCESS – FRONT END LOADING

PHASE	FEL I	FEL II	FEL III	APPROVAL	FEL IV
Project Phase	Planning / Conceptual Design	Feasibility Design	Preliminary Engineering		Detail Engineering
Engineering Definition	2% to 5%	10% to 30%	30% to 60%		60% to 90%
AACE Estimate Class	Class 5	Class 4	Class 3		Class 2
AACE Cost Estimate Accuracy	L: -20% to -50% H: +30% to +100%	L: -15% to -30% H: +20% to +50%	L: -10% to -20% H: +10% to +30%		L: -5% to -15% H: +5% to +20%
Basis of Estimate	Parametric models, factors, experience / judgement	Budgetary quotes, factors, AE estimates	Semi-detailed unit costs, key contract RFPs		Detailed take-offs / unit costs, quoted contracts

Front End Loading (FEL)
Front-End Engineering Design (FEED)

- Used for complex solution deliveries that are typically Engineering to Order (ETO).
- Decision and gate based with cost and ETO uncertainties.
- Examples: Power station, metro, Berlin Airport.

DEVELOPMENT PROCESS – SYSTEMS ENGINEERING



Systems Engineering:

- Development methology defining the complex product systems from multiple viewpoint.
- Requirements, Functional, Logical and Physical product definition enables modularity.
- Integration, verification, validation and qualification ensure the system performs according to requirements.
- Examples: F-35, BMW 5-Series, Ponsse Scorpion

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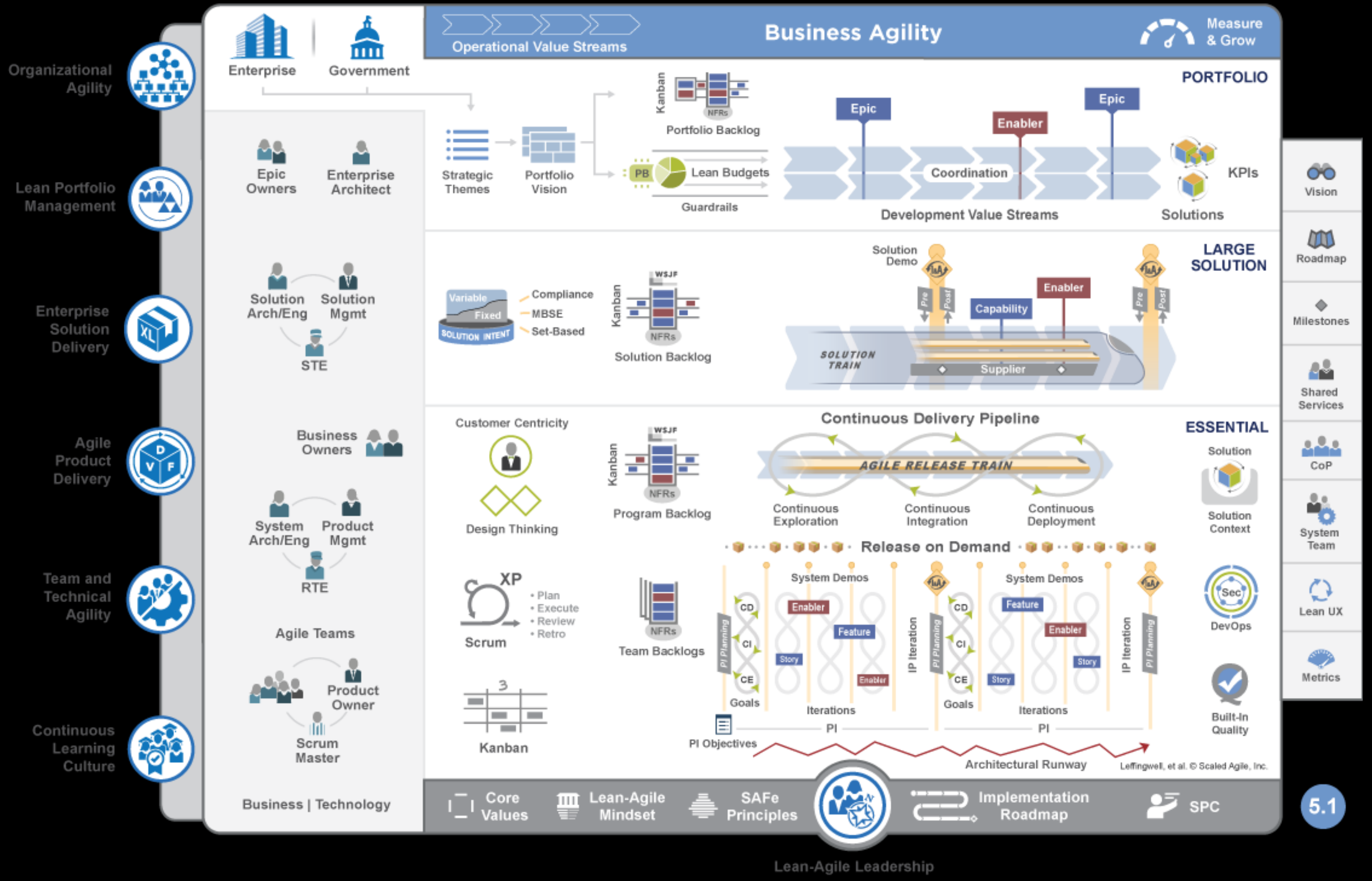
DEVELOPMENT PROCESS – AGILE DEVELOPMENT



Agile Development Model Example

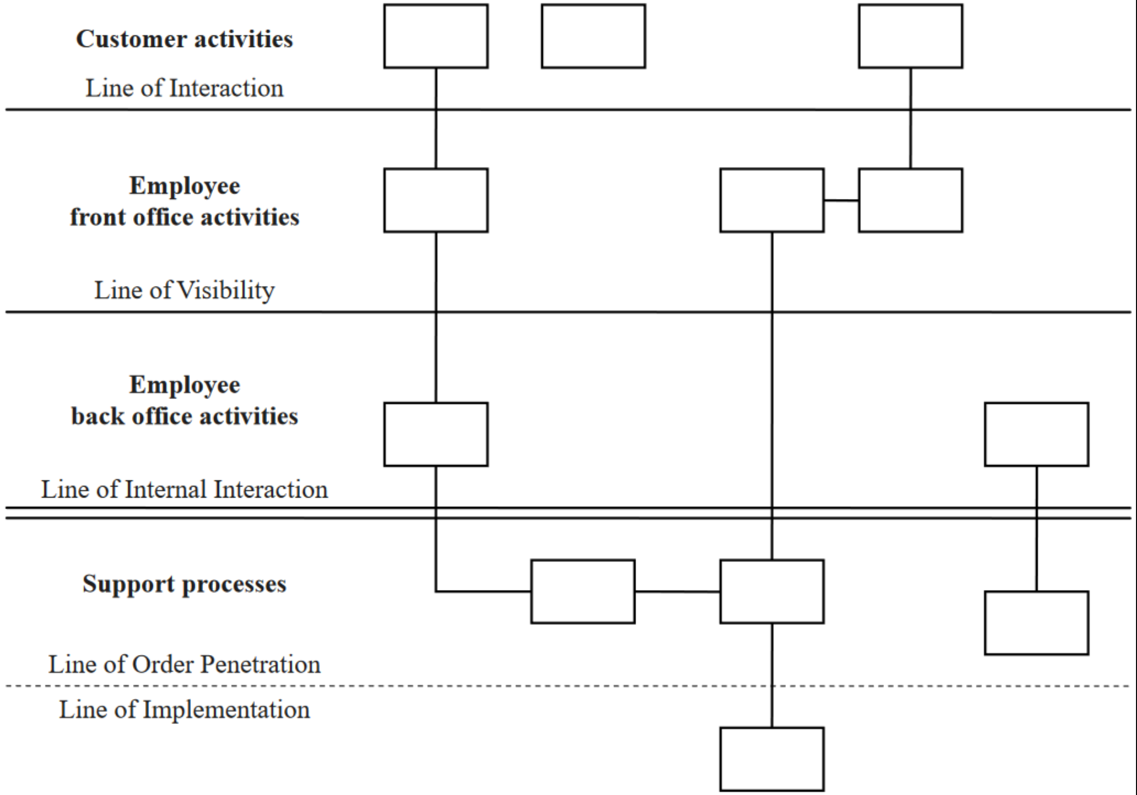
- Focus on agility and feedback from users.
- Originated in the software industry and applied to other industries.
- Agile organisation (ketterä organisaatio).
- SAFe supports multiple different sizes of delivery.
- Examples: Salesforce, iPhone, Kone

DEVELOPMENT PROCESS – SERVICE BLUEPRINT

<table>
<tr><th></th><th>Aware</th><th>Join</th><th>Use</th><th>Develop</th><th>Leave</th></tr>
<tr><td>Product</td><td rowspan="5">The point where the user learns about the services</td><td rowspan="5">The sign-up or registration to the service</td><td rowspan="5">The use of the service</td><td rowspan="5">The user’s expanding usage ot the service</td><td rowspan="5">The point when the user finishes using the service, either for a single session or forever</td></tr>
<tr><td>Customer Actions</td></tr>
<tr><td>Onstage (visible employee actions)</td></tr>
<tr><td>Backstage (invisible employee actions)</td></tr>
<tr><td>Support Process</td></tr>
</table>

- Service Blueprint pioneer G. Lynn for Citibank.
- Combination of understanding the customer journey and interaction of a service.
- Examples: banking service or any digital service.

SERVICE BLUEPRINT



Source: Adapted from Zeithaml and Bitner (1996)

- Interactions along the lifecycle that require front office and back office activities.
- Importance increase when service are digital. CX become critical for service success and revenue.
- Examples: Rolls-Royce Power-by-the-Hour, Foodora, Amazon Web Services, eCommerce, Valmet Safe Harbour.


ANY
QUESTIONS
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